DAIMLER

integra

Guideline for Machinery and Production Equipment: Electrical Equipment

Version

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Purpose of this document

The present Guideline Chapters 5 - 18 contain extensions to the draft of DIN EN 60204-1: 2014. The appendices of the standard must be observed. Chapters 19 - 22 contain addenda to MBN 9666-3.

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Update Service

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Update Journal

No.	Chapter	Page	Contents of modification	V.	Date	Author
1	Initial creation	New version I6	Supplements to MBN 9666-3	60.00	18.11.16	Raschbacher
2	**DC Supplies**	12	Current limit for 24 V power supply units inserted	60.01	22.03.17	Raschbacher
3	11.2.3 Thermal effects	19	Adjust internal switch cabinet temperature if devices with T rating <50° are installed.	60.01	13.11.17	Gaiser
4	13.1 Connections	21	Added note that wire-end sleeves no longer needed	60.01	13.11.17	Gaiser
5	Current-carrying Capacity in Normal Service	20	Definition of the handling of reduction factors according to 60204-1	60.01	13.11.17	Raschbacher
6	M12 Sensor/Actuator Cables	24	Definition of use of M8/12 cables	60.01	15.4.19	Raschbacher
7	General	TOC	Revision of headings in accordance with draft standard 60204-1:2014	60.01	15.4.19	Raschbacher
8	M12 Cables for Connections with Safety Requirements	23	Inserted the AIDA cable and connector color specifications	60.01	02.05.	Raschbacher
9	Connection Boxes and other Boxes	25	Specifications for connector technology in the field and terminal boxes	60.01	02.05.	Raschbacher
10	Ferrulles :	23	Detailed specifications for the use of wire-end sleeves	60.01	02.05.	Raschbacher
11	Single Wire Labeling in Standard Cabinets and Cabinets Made In-house:	22	Detailed specifications for single wire labeling	60.01	02.05.	Raschbacher
12	Identification of the Protective Conductor / Protective Bonding Conductor	23	Explanation of use of gn-ye insulation vs bare conductor	60.01	02.05.	Raschbacher
13	Control Consoles on Kick-in / Kick-out Stations	18	Prescribed procedures for using IO link for control consoles	60.01	02.05.	Raschbacher
14	**DC Supplies**	12	Specifications for setting 24 V power supply unit and for supply of EBF and TBRs	60.01	02.05.	Raschbacher
15	Degrees of Protection	20	Specifications for IP protection class	60.01	02.05.	Raschbacher
15	Enclosures, Doors and Openings	20	EMC specifications for dealing with housing openings	60.01	02.05.	Raschbacher
	General Requirements	21	Specifications for documentation of individual parts for plug connections	60.01	02.05.	Raschbacher
	Conductor and Cable Runs	22	Supplements to gutter construction and reference to guidelines	60.01	02.05.	Raschbacher

The current changes relating to the older version are highlighted in yellow

No. = Serial number **V.** = Edition version (document version, release)

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1 Scope

The General Electrics guideline supplements MBN 9666-3 and contains additional requirements specifically for the assembly, body, paintwork and stamping plants.
The latest version of MBN 9666-3 shall be obtained from the Daimler Supplier Portal and used with this guideline.

The corresponding guidelines in the ECM volumes 1, 2 and 3 remain valid and shall also be taken into account as part of an RFQ. Possible contradictions shall be disclosed and clarified in the engineering design meetings and recorded in writing

Access by suppliers to MBN 9666

We offer our suppliers the possibility to access internal standards, rules & regulations and technical documents of Daimler AG in the DocMaster system. You can access the system online via our supplier portal at:

Daimler Supplier Portal
https://daimler.portal.covisint.com/web/portal

You can register for the portal via the link "REGISTER for the Daimler Supplier Portal".
Once you have accessed the system, you can display, print or save any of the documents in the standard folders (MB Special Terms, CAD Handbook, Logistics Manual).

2 Normative References

3 Terms and Definitions

4 General Requirements

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4.1 General

4.2 Selection of Equipment

4.2.1 General

Only materials that are contained in the material approval lists may be used.

4.2.2 Switchgear

Devices with a cage nut or spring-loaded connection technology are preferred so that wire-end sleeves are not needed.

4.3 Electrical Supply

4.3.1 General

4.3.2 AC Supplies

4.3.3 DC Supplies

The 40 A power supply units in the +H111 station cabinet may only be loaded up to a maximum of 35.00 A with a preset Ua of 27.5 V.

4.3.4 Special Supply Systems

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4.4 Physical Environment and Operating Conditions

4.4.1 General

4.4.2 Electromagnetic Compatibility (EMC)

4.4.3 Ambient Air Temperature

4.4.4 Humidity

4.4.5 Altitude

4.4.6 Contaminants

4.4.7 Ionizing and Non-ionizing Radiation

4.4.8 Vibration, Shock, and Bump

4.5 Transport and Storage

4.6 Provisions for Handling

5 Incoming Supply Conductor Terminations and Devices for Disconnecting and Switching Off

5.1 Incoming Supply Conductor Terminations

5.2 Terminal for Connection to the External Protective Earthing System

5.3 Supply Disconnecting (Isolating) Device

5.3.1 General

5.3.2 Type

5.3.3 Requirements

5.3.4 Operating Means of the Supply Disconnecting Device

5.3.5 Excepted Circuits

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5.4 Devices for Removal of Power for Prevention of Unexpected Start-up

5.5 Devices for Isolating Electrical Equipment

5.6 Protection against Unauthorized, Inadvertent and/or Mistaken Connection

6 Protection against Electric Shock

6.1 General

6.2 Basic Protection ~~Overcurrent Protection~~

6.2.1 General

6.2.2 Protection by Enclosures

6.2.3 Protection by Insulation of Live Parts

6.2.4 Protection against Residual Voltages

6.2.5 Protection by Barriers

6.2.6 Protection by Placing out of Reach or Protection by Obstacles

6.3 Fault Protection

6.3.1 General

6.3.2 Prevention of the Occurrence of a Touch Voltage

6.3.3 Protection by Automatic Disconnection of Supply

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6.4 Protection by the Use of PELV or SELV

7 Protection of Equipment

7.1 General

7.2 Overcurrent Protection

7.2.1 General

7.2.2 Supply Conductors

7.2.3 Power Circuits

7.2.4 Control Circuits

7.2.5 Socket- Outlets and their Associated Conductors

7.2.6 Lighting Circuits

7.2.7 Transformers

7.2.8 Location of Overcurrent Protective Devices

7.2.9 Overcurrent Protective Devices

7.2.10 Rating and Setting of Overcurrent Protective Devices

7.3 Protection of Motors against Overheating

7.3.1 General

7.3.2 Overload Protection

7.3.3 Over-temperature Protection

7.4 Protection against Abnormal Temperature

7.5 Protection against the Effects of Supply Interruption or Voltage Reduction and Subsequent Restoration

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7.6 Motor Overspeed Protection

7.7 Additional Earth Fault/Residual Current Protection

7.8 Phase Sequence Protection

7.9 Protection against Overvoltages Due to Lightning and Switching Surges

7.10 Short Circuit Current Rating

8 Equipotential Bonding

8.1 General

The equipotential bonding system is carried out according to 000_B3.26.05.03_GL_grounding Concept_line Engineering_Vxx.yy.

It is not allowed to install devices with voltage levels > 50V AC/DC ath the protecting fence

8.2 Protective Bonding Circuit System

8.2.1 General

8.2.2 Protective Conductors

8.2.3 Continuity of the Protective Bonding Circuit

8.2.4 Protective Conductor Connecting Points

8.2.5 Mobile Machines

8.2.6 Additional Requirements for Electrical Equipment Having Earth Leakage Currents Higher than 10 mA

8.3 Measures to Restrict the Effects of High Leakage Current

8.4 Functional Bonding

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9 Control Circuits

9.1 Control Circuits

9.1.1 Control Circuit Supply

9.1.2 Control Circuit Voltages

9.1.3 Protection

9.2 Control Functions

9.2.1 Categories of Stop Functions

9.2.2 Operation

9.2.3 Cableless Control

9.3 Protective Interlocks

9.3.1 Reclosing or Resetting of an Interlocking Safeguard

9.3.2 Exceeding Operating Limits

9.3.3 Operation of Auxiliary Functions

9.3.4 Interlocks between Different Operations and for Contrary Motions

9.3.5 Reverse Current Braking

9.3.6 Suspension of Safety Functions and/or Protective Measures

9.4 Control Functions in the Event of Failure

9.4.1 General Requirements

9.4.2 Measures to Minimize Risk in the Event of Failure

9.4.3 Protection against Maloperation Due to Insulation Faults, Voltage Interruptions and Loss of Circuit Continuity

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10 Operator Interface and Machine-Mounted Control Devices

10.1 General

Control Consoles on Kick-in / Kick-out Stations

Pluggable IO link control consoles and associated lamps are to be used at kick-in / kick-out stations with a maximum of 6 variants, see 000_B3 026 04 02_HVO_Integra_PN_EN_Vxx.yy.

10.1.1 General Requirements

10.1.2 Location and Mounting

10.1.3 Protection

10.1.4 Position Sensors

10.1.5 Portable and Pendant Control Stations

10.2 Actuators

10.2.1 Colours

10.2.2 Markings

10.3 Indicator Lights and Displays

10.3.1 General

10.3.2 Colours

10.3.3 Flashing Lights and Displays

10.4 Illuminated Push-buttons

10.5 Rotary Control Devices

10.6 Start Devices

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10.7 Emergency Stop Devices

10.7.1 Location of Emergency Stop Devices

10.7.2 Types of Emergency Stop Device

10.7.3 Operation of the Supply Disconnecting Device to Effect Emergency Stop

10.8 Emergency Switching Off Devices

10.8.1 Location of Emergency Switching Off Devices

10.8.2 Types of Emergency Switching Off Device

10.8.3 Local Operation of the Supply Disconnecting Device to Effect Emergency Switching Off

10.9 Enabling Control Device

11 Controlgear: Location, Mounting, and Enclosures

11.1 General Requirements

11.2 Location and Mounting

11.2.1 Accessibility and Maintenance

11.2.2 Physical Separation or Grouping

11.2.3 Heating Effects

A heat calculation (000_B3.02.04.30.05_APP_calculation_airconditioning_drive cabinets_Vxx.yy) must be submitted for frequency converter control cabinets built according to the EPLAN sample template and the necessary air conditioning unit must be selected according to the required cooling capacity.
If components in the switch cabinet have a temperature resistance lower than 50 degrees, the Ti value (switch cabinet interior temperature) must be adjusted accordingly.

An ambient temperature Tu of 40 degrees should generally be used to calculate the power dissipation in the switch cabinet.
The theoretical internal temperature for the dT calculation must be assumed to be Ti 50 degrees.
During commissioning, the switching point of the temperature control must be set to 40 degrees.

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When calculating the resulting power loss, the simultaneity of the installed components such as frequency converters must be taken into account.

Special switch cabinet locations:
For locations like e.g. Heating cellar, shed roof, painting (heating) process environments, the ambient temperature Tu must be discussed and determined as part of project negotiations.

11.3 Degrees of Protection

The field installations installed by the system installer, in particular the decentralized I/O and bus systems, must at least meet the IP 65 ingress protection level.

11.4 Enclosures, Doors and Openings

The cable glands from below shall be sealed (base plate using foam rubber).
EMC shielding (earthing connections) must be connected. Door seals must be provided with a conductive coating.

In switch cabinets with increased EMC requirements, wall bushings compatible with EMC requirements which offer tightness requirements and strain relief can also be used for cable entry.

11.5 Access to Electrical Equipment

12 Conductors and Cables

12.1 General Requirements

The release for a technically comparable cable can be obtained from the Daimler technical planner in the event of a deviation from the control and power cables listed in the approval list.

12.2 Conductors

12.3 Insulation

12.4 Current-carrying Capacity in Normal Service

DIN EN 60204-1 prescribes reduction factors for conductors under steady-state conditions for different methods of installation. These can be determined from Table 6.
The reduction in the maximum cable load capacity can be dispensed with in the case of modules whose power requirement assessment, taking into consideration simultaneity factors, is <= 50% of the rated current. The circuit-breaker should be in accordance with Guidelines 000_B3.026.05.04-DE-EN_Table_Technology_Connection_Vxx.yy....

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12.5 Conductor and Cable Voltage Drop

12.6 Flexible Cables

12.6.1 General

12.6.2 Mechanical Rating

12.6.3 Current-carrying Capacity of Cables Wound on Drums

12.7 Conductor Wires, Conductor Bars and Slip-ring Assemblies

12.7.1 Basic Protection

12.7.2 Protective Conductor Circuit

12.7.3 Protective Conductor Current Collectors

12.7.4 Removable Current Collectors with a Disconnector Function

12.7.5 Clearances in Air

12.7.6 Creepage Distances

12.7.7 Conductor System Sectioning

12.7.8 Construction and Installation of Conductor Wire, Conductor Bar Systems and Slip-ring Assemblies

13 Wiring Practices

13.1 Connections and Routing

13.1.1 General Requirements

Screened cables shall be connected directly to the component in the switch cabinet,
i.e. clamps are not used. The screen shall additionally be grounded at the cabinet entrance (EMC cable clamping rail, or EMC screw connection).
The manufacturer's instructions regarding the connection of the screen shall be observed.

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Each complete plug connection (grommet and connector housing, inserts, pins and sockets) of all the plug-in connections installed in the system and in the switchgear cabinets that have been delivered are part of the scope of delivery. The individual parts of the plug connections must be listed in the electrical documentation.

On the terminal strips, the connections having a common voltage shall be grouped together as much as is feasible. The spatial assignment of the equipment grounding conductor connection terminals to the control line terminals shall be adhered to.

13.1.2 Conductor and Cable Runs

See Guidelines 000_B3.047.01_GL-EN_Media Installation_Vxx. yy
Floor mounted cable ducts in the system must be of a walk-on design according to the material approval list.

Cable apertures and sharp edges shall have edge protection, and cable bushings shall have cable protecting sleeves.

The cable duct covers shall have a fully closed design and be safe to walk on at the intersections.

In areas with foot traffic, cable ducts shall be provided with grooved sheet metal that is easy to remove. The fastening can be using steel springs or twist bolts and must be coordinated with the responsible department in the project negotiations.

The cable ducts shall be installed in the systems so that ground-level access to the maintenance areas is ensured. Where this is not possible, the cable duct must be designed as a maintenance path (for width, see GL Safety Engineering).

The cable trays, cable channels and cable ducts shall be dimensioned in coordination with the responsible Daimler department so that enough space is still available for additional installations of the building services, supply technology and communication technology.

Vertically mounted cable channels up to a height of 2.5 m must be closed with hinged lids. Cables and conductors must be properly fastened with straps.

13.1.3 Conductors of Different Circuits

13.1.4 Connection between Pick-up and Pick-up Converter of an Inductive Power Supply System

13.2 Identification of Conductors

13.2.1 General Requirements

Single Wire Labeling in Standard Cabinets and Cabinets Made In-house:

If wire routing with overprinted labeling is used for internal wiring, the additional individual wire labeling can be dispensed with.

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In special switch cabinets or when rewiring without wire routing, individual wire labeling must be applied if the cable routing to the switching device is not TRACEABLE, e.g. for DIN rail-mounted control devices with individual connections. If assignment to the terminals of the device is possible by means of the Lütze comb, the individual wire marking can be dispensed with.

In the USA, single wire labeling must be used for switchgear of this kind if wire routing is not used.

13.2.2 Identification of the Protective Conductor / Protective Bonding Conductor

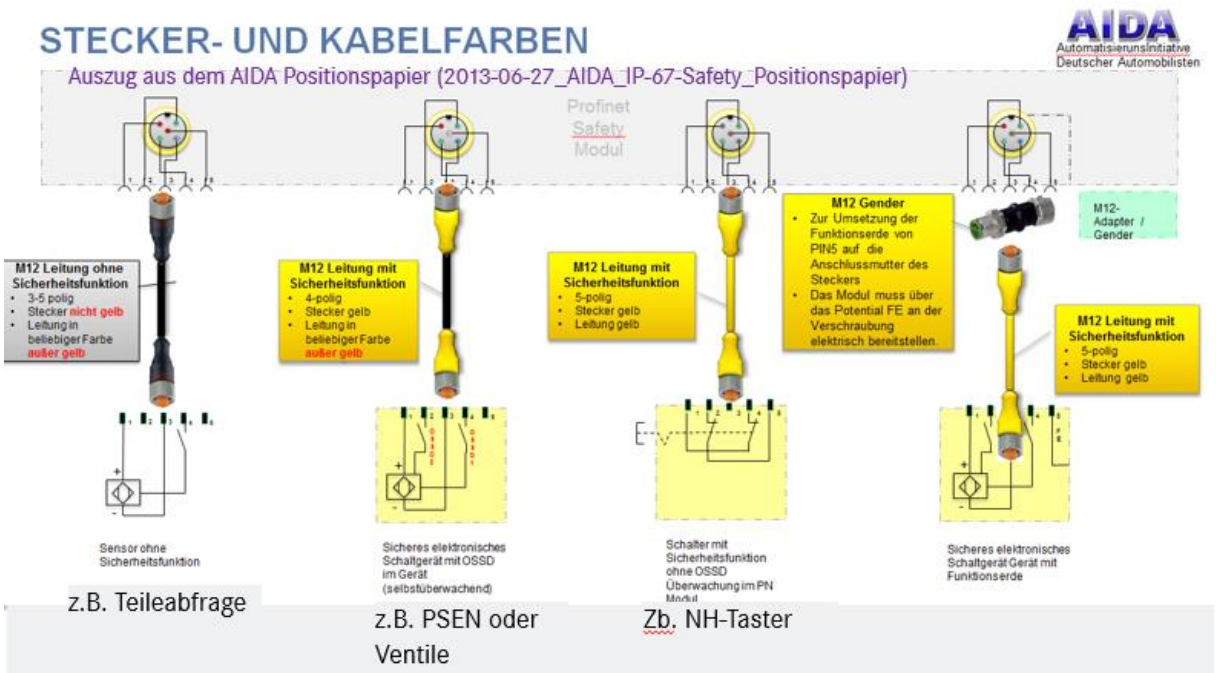
Within the switch cabinet, the protective conductor is designed with gn-ye insulation to ensure dielectric strength. Functional equipotential bonding conductors to side walls or assembly frames should be designed with EMC ribbon cables. The cables should consist of tinned copper wires.

Outside the switch cabinet, the equipotential bonding conductors are uninsulated bare conductors or gn-ye insulated conductors in drag chains or dress packages. Labeling is carried out by marking the connection point on the device/component.

13.2.3 Identification of the Neutral Conductor

13.2.4 Identification by Colour

M12 Cables for Connections with Safety Requirements



13.3 Wiring Inside Enclosures

Ferrulles :

Ferrulles are not used for spring-loaded and cage clamps unless the manufacturer of the terminal specifies the use of ferrulles. Where appropriate, the prescribed crimping form must also be observed.

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No ferrulles are required with the cage terminals of the N/PE potential rails.

For terminals with screw terminals without a cage, where the clamping screw presses directly on the strands, ferrulles are still mandatory.

Ferrulles must also be used on the Lenze frequency converters for cross-sections <1 mm², and if prescribed by the manufacturer of the device.

On the connection sockets of the ET200SP modules, ferrulles should be used for cross-sections of flexible cables <= 0.5 mm² in order to facilitate installation.

The uninsulated tinned 10 mm² equipotential bonding conductors which are laid outside the switch cabinet between the earthing cable and the connection point must have ferrulles crimped onto them.

13.4 Wiring Outside Enclosures

13.4.1 General Requirements

M12 Sensor/Actuator Cables

M12 cables with LEDs are required wherever the switchgear has no visible diagnostic LEDs. The diagnostic LEDs in the cable must show the power supply to pin 1 (for connection to sensors), pin 2 and pin 4.
M12 cables with LEDs must always be connected to sensors with M12 pigtails so that defective pigtails can be detected.
M12 pigtails on sensors must always be installed in such a way that the LEDs of the connected M12 cable are visible.
The M12 sensor/actuator cables should point downwards when connected to the sensor/actuator in order to apply as little torsion as possible to the plug connection. Depending on the application, this must be done with straight or angled plugs.
Intermediate plugs to extend cables are not permitted. Plug connections must not be concealed during installation. Cable length must not exceed 15 m, otherwise an additional I/O module must be installed.

Y-distributors must be equipped with LEDs.

Sensors/actuators with M8 connection technology should be avoided. They must be converted to M12 with the cable adapter as soon as possible.

13.4.2 External Ducts

13.4.3 Connection to Moving Elements of the Machine

The cables in all cable trailing devices shall be connectable at both bends.

The lines or media in cable trailing device connections shall be separated using separators and individually strain-relieved at the fixed point and driver.

Velcro strips shall be used to bundle cables and lines for the entire machine installation.

Attachments with self-locking strain relief (cable wedge clamps) shall be used in the case of vertical cable trailing devices.

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13.4.4 Interconnection of Devices on the Machine

13.4.5 Plug/Socket Combinations

13.4.6 Dismantling for Shipment

13.4.7 Additional Conductors

13.5 Ducts, Connection Boxes and Other Boxes

13.5.1 General Requirements

13.5.2 Percentage Fill of Ducts

An additional reserve space of 20% shall be available on the cable platform and in cable trays and ducts next to the reserved final status (e.g. reserve stations, derivatives …).
The cable platform and cable trays and channels shall be dimensioned in coordination with the Daimler project management in such a way that sufficient space is still available for additional installations of facilities, supply technology and communication technology.

13.5.3 Rigid Metal Conduit and Fittings

13.5.4 Flexible Metal Conduit and Fittings

13.5.5 Flexible Non-Metallic Conduit and Fittings

13.5.6 Cable Trunking Systems

13.5.7 Machine Compartments and Cable Trunking Systems

13.5.8 Connection Boxes and other Boxes

The use of terminal boxes is only intended for the installation of decentralized IP20 network switches. Decentralized IP67 modules of the approved products are to be used to control the sensors/actuators in the field.

Plug-in connection technology

The electrical installation in the field is to be carried out as far as possible with plug-in connectors. Cable glands and direct wiring in terminal boxes and control panels should be avoided.
M12 A-coded connectors are used for the I/O system. AIDA PushPull connectors are used for the 24 V power supply and Profinet.

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Safe M12 interfaces

M12 interfaces for the transmission of safety-related signals must use M12 panel feed-throughs with 4 and 5 poles and yellow insulation inserts.

13.5.9 Motor Connection Boxes

14 Electric Motors and Associated Equipment

14.1 General Requirements

14.2 Motor Enclosures

14.3 Motor Dimensions

14.4 Motor Mounting and Compartments

14.5 Criteria for Motor Selection

14.6 Protective Devices for Mechanical Brakes

15 Socket-Outlets and Lighting

15.1 Socket-outlets for Accessories

15.2 Local Lighting of the Machine and of the Equipment

15.2.1 General

The area lighting of the switchgear cabinets shall be positioned such that sufficient lighting is provided when working on an opened switchgear cabinet and internal switchgear cabinet lighting is not required.

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15.2.2 Supply

15.2.3 Protection

15.2.4 Fittings

16 Marking, Warning Signs and Reference Designations

16.1 General

16.2 Warning Signs

16.2.1 Electric Shock Hazard

16.2.2 Hot Surfaces Hazard

16.3 Functional Identification

16.4 Marking of Enclosures of Electrical Equipment

16.5 Reference Designations

17 Technical Documentation

17.1 General

17.2 Information Related to the Electrical Equipment

18 Verification

18.1 General

18.2 Verification of Conditions for Protection by Automatic Disconnection of Supply

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18.2.1 General

18.2.2 Test 1 – Verification of the Continuity of the Protective Bonding Circuit

18.2.3 Test 2 – Fault Loop Impedance Verification and Suitability of the Associated Overcurrent Protective Device

The correct design of the electric circuits with regard to the fuse dimensioning shall be verified via short-circuit simulation. The impedance values have been calculated and shall be verified via measurement.

18.2.4 Application of the Test Methods for TN Systems

18.3 Insulation Resistance Tests

18.4 Voltage Tests

18.5 Protection against Residual Voltages

18.6 Functional Tests

18.7 Retesting

18.7.1 Dimensioning of Fuses

19 Electronic Equipment

19.1 General

19.2 Basic Requirements

19.2.1 Memory Capacities

19.2.2 Inputs and Outputs

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19.2.3 Diagnostic Messages

19.2.4 Measuring Systems

20 Programmable Equipment

20.1 Programmable Controls

20.2 Backup Battery and Protection

20.3 Software Check

21 Commissioning, Technical Acceptance, Training

21.1 Commissioning

21.2 Technical Acceptance

21.2.1 General

21.2.2 Conditions for Acceptance

21.2.3 Handover of Technical Documents

21.3 Training

22 Measures to Facilitate the Efficient Use of Energy

22.1 Sleep Mode

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22.2 Heat / Cooling Supply

23 Component Release List

The MDM (Material Data Manager) shall no longer be used for new systems in the assembly, body, paintwork and stamping plants. Consultation with project management - including the involvement of the responsible Daimler department - is required in the case of converted systems (re-use projects).

Use of the material approval lists is described in Volume 3.10.01 "Material Approval Lists"

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